

Problem Set 2 — Vector Calculus

VCE Specialist Mathematics 3&4 · Chapter 13 · Week 2

Due: start of Week 3. Answers at the back — attempt everything first.

Name: _____

Due: Monday 27 July

Method reminder: calculus acts componentwise on $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. Velocity $\mathbf{v} = \dot{\mathbf{r}}$ is tangent to the path; speed = $|\mathbf{v}|$; collisions need equal position at equal time; for closest approach minimise d^2 .

Section A — Paths and derivatives

A1. A particle has position $\mathbf{r}(t) = 3t\mathbf{i} + (4t - t^2)\mathbf{j}$ (metres, seconds), $t \geq 0$.

- a.) Find $\mathbf{v}(t)$ and $\mathbf{a}(t)$. *(2 marks)*
- b.) Find the initial speed. *(2 marks)*
- c.) Find the Cartesian equation of the path. *(2 marks)*

A2. A particle moves on the path $\mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j}$.

- a.) Describe the path (shape, centre, radius, direction). *(2 marks)*
- b.) Show the speed is constant, and find it. *(2 marks)*
- c.) Show that $\mathbf{a}(t) = -\mathbf{r}(t)$, and interpret the direction of the acceleration. *(2 marks)*

Section B — Motion in the plane

B1. $\mathbf{r}(t) = (t^2 - 4t)\mathbf{i} + 2t\mathbf{j}$, $t \geq 0$.

- a.) Find when and where the particle crosses the y -axis. *(3 marks)*
 - b.) Find $\mathbf{v}(t)$, and the time at which the speed is a minimum. *(3 marks)*
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B2. A projectile is launched from the origin with $\dot{\mathbf{r}}(0) = 15\mathbf{i} + 20\mathbf{j}$ m/s, subject to $\ddot{\mathbf{r}} = -10\mathbf{j}$ m/s².

a.) Find $\mathbf{r}(t)$. *(3 marks)*

b.) Find the greatest height reached, and the time at which it occurs. *(2 marks)*

c.) Find where the projectile lands ($y = 0$), and its speed on landing. *(3 marks)*

B3. Particles P and Q have positions $\mathbf{r}_P(t) = (t + 1)\mathbf{i} + t^2\mathbf{j}$ and $\mathbf{r}_Q(t) = 3\mathbf{i} + 2t\mathbf{j}$. Show that P and Q collide, and find the time and place of the collision. *(4 marks)*

Section C — Closest approach and distance along a path

C1. A ship's position relative to a lighthouse at the origin is $\mathbf{r}(t) = (2t - 6)\mathbf{i} + t\mathbf{j}$ km after t hours, $t \geq 0$.

a.) Find the time at which the ship is closest to the lighthouse. *(3 marks)*

b.) Find this minimum distance, in exact form. *(2 marks)*

C2. For the particle $\mathbf{r}(t) = 5 \cos t \mathbf{i} + 5 \sin t \mathbf{j}$, find the distance travelled from $t = 0$ to $t = \frac{\pi}{2}$, and check it against the geometry of the path. *(3 marks)*

C3. A particle has $\mathbf{r}(t) = t \mathbf{i} + t^2 \mathbf{j}$, $0 \leq t \leq 1$. Write the distance travelled as a definite integral, and evaluate it with your CAS to three decimal places. *(3 marks)*

Answers

A1. a.) $\mathbf{v} = 3\mathbf{i} + (4 - 2t)\mathbf{j}$; $\mathbf{a} = -2\mathbf{j}$. b.) $|\mathbf{v}(0)| = \sqrt{9 + 16} = 5$ m/s. c.) $y = \frac{4x}{3} - \frac{x^2}{9}$.

A2. a.) Circle, centre O , radius 2, anticlockwise from $(2, 0)$. b.) $|\mathbf{v}| = \sqrt{4\sin^2 t + 4\cos^2 t} = 2$. c.) $\mathbf{a} = -2\cos t\mathbf{i} - 2\sin t\mathbf{j} = -\mathbf{r}$: acceleration points to the centre.

B1. a.) $t^2 = 4t \Rightarrow t = 0$ (at O) and $t = 4$, at $(0, 8)$. b.) $\mathbf{v} = (2t - 4)\mathbf{i} + 2\mathbf{j}$; $\text{speed}^2 = (2t - 4)^2 + 4$ minimised at $t = 2$ (speed 2 m/s).

B2. a.) $\mathbf{r} = 15t\mathbf{i} + (20t - 5t^2)\mathbf{j}$. b.) 20 m at $t = 2$ s. c.) Lands at $t = 4$: $(60, 0)$; $\mathbf{v}(4) = 15\mathbf{i} - 20\mathbf{j}$, speed 25 m/s.

B3. i: $t + 1 = 3 \Rightarrow t = 2$; $\mathbf{j}$ at $t = 2$: $t^2 = 4 = 2t$ ✓. Collide at $(3, 4)$ when $t = 2$.

C1. a.) $d^2 = (2t - 6)^2 + t^2 = 5t^2 - 24t + 36$; $\frac{d}{dt}(d^2) = 10t - 24 = 0 \Rightarrow t = 2.4$ h. b.) $d^2 = 7.2 \Rightarrow d = \frac{6}{\sqrt{5}} = \frac{6\sqrt{5}}{5} \approx 2.68$ km.

C2. Speed = 5, so distance = $5 \cdot \frac{\pi}{2} = \frac{5\pi}{2}$ — a quarter of a circle of circumference 10π . ✓

C3. $\int_0^1 \sqrt{1 + 4t^2} dt \approx 1.479$.